

Cellulose nanofiber-based hydrogels with high mechanical strength

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Abstract The preparation of high-strength hydrogels from plant-based cellulose nanofibers by simple alkaline treatment is described herein. We isolated the cellulose nanofibers with a uniform width of approximately 15 nm from wood and we prepared two types of hydrogel sheet with different crystal forms (celluloses I and II) in 9 and 15 wt% aqueous sodium hydroxide solutions. Both of the hydrogels exhibited high tensile properties because of the crystalline network in the gels. Especially, the nanofiber hydrogel with a cellulose II crystal structure with the swelling degree of 13.4 achieved a Young's modulus and tensile strength in excess of 35 and 5 MPa respectively, because it had a continuous and strong nano-network formed via the interdigitation of the neighboring nanofibers during mercerization.

Keywords Cellulose nanofiber · Hydrogel · Alkaline treatment · Mercerization · Tensile properties

Introduction

Plant cell walls consist of a complex gel structure with rigid cellulose microfibrils embedded in a hydrated matrix of polysaccharides such as hemicellulose and

pectins. Such natural polysaccharides have been widely used for the preparation of hydrogels in various fields in, for example, biomedical, pharmaceutical and tissue engineering applications because of their many advantages, including their biocompatibility biodegradability and low toxicity (Coviello et al. 2007). Cellulose, the most abundant renewable polysaccharide on earth, is a strong candidate for the fabrication of hydrogels and many ingenious designs of cellulose-based hydrogels have been reported including cellulose-polymer composite hydrogels and cellulose-inorganic hybrid hydrogels (Chang and Zhang 2011). Cellulose, which is a long linear polymer of D-glucose with three free hydroxyl groups, is insoluble in water, and thus the molecular chains synthesized on the plasma membrane are highly crystallized into microfibrils with diameters of ~3 nm in higher plant cell walls (Somerville et al. 2004). Since this crystalline structure prevents it from dissolving even in common organic solvents, the preparation of cellulose-based hydrogels almost always involves the synthesis of water-soluble cellulose derivatives or the use of specific solvents such as LiCl/dimethylacetamide, *N*-methylmorpholine-*N*-oxide, ionic liquids and alkali/urea aqueous systems.

On the other hand, the stable crystalline structure provides the microfibrils with excellent longitudinal mechanical properties, including high Young's modulus and very low coefficient of thermal expansion. Therefore, the nano-sized microfibrils, so-called cellulose nanofibers, have been receiving much attention

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as a strong candidate for effective reinforcement of polymers. Recently the isolation from various plant sources can be attained by mechanical and/or chemical methods (Eichhorn et al. 2010; Siro and Plackett 2010). We have isolated cellulose nanofibers from wood, rice straw, potato tubers and bamboo using a grinder (Abe et al. 2007; Abe and Yano 2009, 2010). The resultant fibers have a uniform width of approximately 15 nm and a length of more than several micrometers and correspond to cellulose microfibril aggregates in the cell walls (Donaldson 2007).

Cellulose nanofibers, which are insoluble in water as mentioned above, homogeneously dispersed in water and the suspension exhibited a high viscosity even with 1 wt% nanofiber because of the large hydrophilic surface area and entanglement. By taking advantage of the unique nature of the nanofiber suspension, we reported the easy preparation of hydrogels from cellulose nanofibers through alkaline treatment (Abe and Yano 2011). Unlike traditional polymer hydrogels, the nanofiber hydrogels could be prepared without dissolving in specific solvents and consisted of a highly porous and crystalline nano-network capable of having two different kinds of crystal form of cellulose I and II in response to the increasing concentration of alkaline solutions.

In our previous study, we described the preparation methods and structural features of nanofiber hydrogels and but the mechanical properties of these hydrogels are not known. However, it is of interest to examine how the crystalline network of nanofiber hydrogel influences their mechanical properties. In the present study, we prepared two types of nanofibers hydrogel sheets with cellulose I and II crystal forms based on our previous work and we focused on evaluating the tensile properties because cellulose nanofibers provide strong resistance to longitudinal tensile stress.

Experimental sections

Preparation of cellulose nanofibers

Wood powder from Radiata Pine (*Pinus radiata*) sieved under 100 mesh was purified to prepare cellulose nanofibers. The sample was dewaxed in a Soxhlet apparatus with a 2:1 (v/v) mixture of toluene/ethanol for 6 h and then air-dried at room temperature. The subsequent procedure followed the method

described in our previous studies (Abe et al. 2007; Abe and Yano 2009, 2010, 2011). In brief, lignin was removed using an acidified sodium chlorite solution and then hemicelluloses were removed using an alkaline treatment with potassium hydroxide. After each chemical treatment, the sample was filtered and rinsed with distilled water until the residues were neutralized. The α -cellulose content of the purified sample was approximately 85 % when determined by extraction with 17.5 wt% sodium hydroxide (NaOH). Lastly, the aqueous suspension with 1 wt% undried purified sample was passed once through a grinder (MKCA6-3; Masuko Sangyo Co., Saitama, Japan) at 1,500 rpm. The grinding treatment was performed with a clearance gauge of -2.5 (corresponding to a 0.25 mm shift) from a zero position.

Preparation of hydrogel sheets

The nanofiber suspension prepared as described above was diluted to 0.1 wt% by distilled water and then vacuum-filtered and dewatered using a polytetrafluoroethylene membrane filter (0.1 μ m mesh). The wet sheets obtained at a fiber content of approximately 10 wt% were immersed in NaOH solutions with concentrations of 9 and 15 wt% at 50 °C for 12 h. The reduction in NaOH concentration caused by immersing the wet sheets was less than 0.1 wt%. Afterward, the sheets were neutralized in 2 % dilute acetic acid for 2 h and then in distilled water for 12 h, at a constant 50 °C. For the resultant hydrogels, the swelling degree (q), defined here as the ratio of the swollen gel mass to that of the dry gel, was calculated as shown in Table 1.

Characterization

X-ray diffraction measurements were carried out with a Rigaku X-ray diffractometer (UltraX 18HF; Rigaku Corp., Tokyo) with CuK α radiation (40 kV and 300 mA) from 5° to 40° in reflection mode. The samples for X-ray diffraction were prepared by hot-pressing at 120 °C into dried sheets.

Test specimens of 30-mm length and 6-mm width were cut from the hydrogel sheets and subjected to a tensile test using a universal materials testing machine (model 3365; Instron Corp., Canton, MA) at a crosshead speed of 10 mm/min with a gauge length of 20 mm. The average values of the Young's modulus,

Table 1 Average tensile properties of cellulose nanofiber hydrogels

	q	E (MPa)	σ (MPa)	ε (%)
9 wt% NaOH	12.1	21.2 (0.4)	3.0 (0.1)	17.5 (0.9)
15 wt% NaOH	13.4	35.2 (0.6)	5.4 (0.2)	21.1 (0.4)

q, Swelling degree; E, Young's modulus; σ , tensile strength; ε , fracture strain

Standard deviation are given in parenthesis

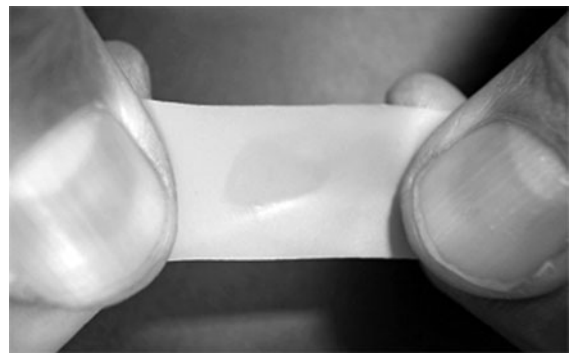
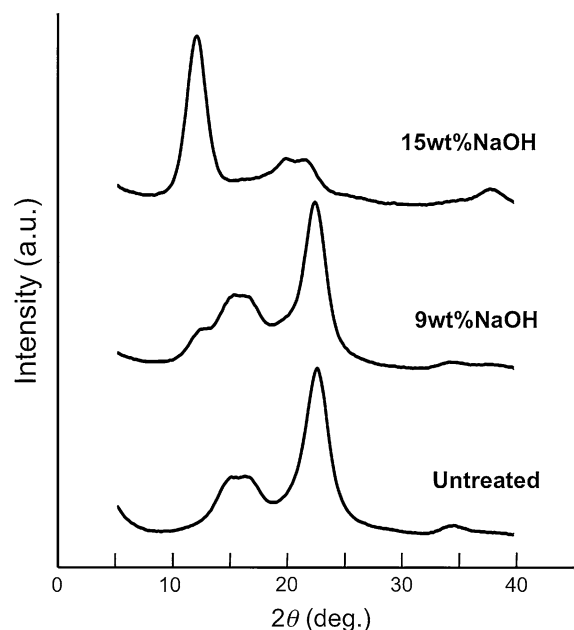
the tensile strength and the fracture strain were calculated for five specimens.

For microscopic observations, small pieces of the hydrogel sheets were slowly dehydrated through an ethanol series (50, 60, 70, 80, 90, 100, 100 %, 1 h each) followed by acetone (1 h) that was then replaced by t-butyl alcohol (1 h), and finally freeze-dried. The freeze-dried samples were coated with platinum by an ion sputter coater and then observed with a field-emission scanning electron microscope (FE-SEM, JSM-6700F; JEOL, Tokyo) operating at 1.5 kV from both the top and side views. Although the coating thickness was approximately 2 nm in this condition, we confirmed that the coating did not change the lateral dimension of fibrils significantly because the ions are sputtered from the top.

Results and discussion

Cellulose nanofibers, which were dispersed in water without sedimentation even at 0.1 wt%, were gently deposited by vacuum filtration and were formed into wet sheets with a fiber content of approximately 10 wt%. Though the wet sheets had an apparently gel-like texture, they were easily disrupted by brief agitation in water, gradually redispersing into the original suspension. This means that the wet sheets had no physical crosslinking such as hydrogen bonding between the nanofibers. However, by soaking in NaOH aqueous solutions and then neutralization, the sheets were formed into stable and tough hydrogels as shown in Fig. 1. These gel formations showed diameter shrinkage of 6.8 and 21.6 %, respectively when prepared in 9 and 15 wt% NaOH, respectively. The significant shrinkage was similar to that seen in our previous study (Abe and Yano 2011).

Studies have shown that strong alkaline treatments change the crystalline structure of native cellulose. In our previous study we found that the alkaline

**Fig. 1** Appearance of the nanofiber hydrogels prepared in 15 wt% NaOH**Fig. 2** Equatorial X-ray diffraction profiles of the dry matters of the nanofiber hydrogels prepared in 9 and 15 wt% NaOH

treatment below and above approximately 12 wt% NaOH resulted in two types of hydrogels with different crystal forms, celluloses I and II (Abe and Yano 2011). Similarly, Fig. 2 shows that the dry

matters of the hydrogels prepared in 9 and 15 wt% NaOH showed mainly the X-ray patterns characteristic of cellulose I and cellulose II crystal forms, respectively. Hereafter we refer to these hydrogels as the Cell-I gel and Cell-II gel. Both hydrogels—especially the Cell-II gel—showed high crystallinity. The crystallinity of the Cell-II gel in this study was significantly higher than those of regenerated cellulose hydrogels or aerogels reported in the literature; for example, when prepared in alkali/urea aqueous solution (Cai et al. 2008) or LiCl/DMSO solution (Wang et al. 2012). Because the Cell-II gel was prepared in 15 wt% NaOH solution at 50 °C without agitation, the cellulose nanofibers were little dissolved in the NaOH solution and were highly recrystallized into cellulose II with less disorder of the cellulose chain (Fig. 2). This suggests that a mercerization with a solid-state process triggered the formation of the Cell-II gel.

Although in the present study the X-ray diffraction measurement was performed on dried samples, the Cell-II gel before drying will probably have the crystal structure of the cellulose II hydrate with similarly high crystallinity. Wada et al. prepared a cellulose II hydrate from microcrystalline cellulose by mercerization followed only by washing with water (Wada et al. 2010). The Cell-I gel also had high cellulose crystallinity but it was slightly decreased compared to the untreated sample, and the profile showed a slight peak at a 2θ angle of approximately 12.3° , which seems to be specific to the $1\bar{1}0$ reflection of cellulose II. These results indicate that both of the Cell-I and Cell-II gels have the unique distinction of being composed of a highly crystalline framework.

Figure 3 and Table 1 show the tensile stress–strain curves and averages of the Young's modulus, tensile strength and fracture strain value of the Cell-I and Cell-II gels. Although the untreated wet sheet was too fragile for the tensile test, both of the gels showed superior tensile performance. The Cell-II gel exhibited higher tensile properties than the Cell-I gel and achieved a Young's modulus and tensile strength in excess of 35 and 5 MPa, respectively. Nakayama et al. developed a high-strength double-network gel using bacterial cellulose and gelatin (Nakayama et al. 2004). The double-network gel with the swelling degree of 3.1 had a Young's modulus of 21 and a tensile strength of 3.8, which are comparable to those of our gels with the swelling degree of 12–13.

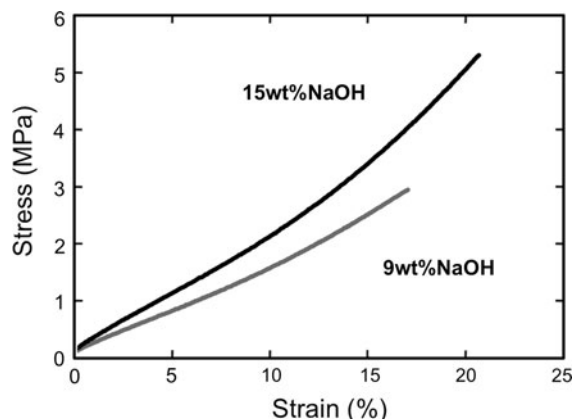


Fig. 3 Tensile stress–strain curves of the nanofiber hydrogels prepared in 9 and 15 wt% NaOH

It seems reasonable to assume that the high tensile properties of the cellulose nanofiber hydrogels can be attributed to the crystalline structure in the gels. However, Nishino et al. measured the elastic modulus of the crystalline regions of cellulose I and cellulose II in the direction parallel to the chain axis as 138 and 88 GPa, respectively (Nishino et al. 1995). The elastic modulus of cellulose II hydrate probably is lower than that of cellulose II because of an increase in the cross-sectional area of the crystal and therefore one might wonder why our Cell-II gel was tougher than the Cell-I gel under tension. The top-view observation in Fig. 4 demonstrates the change in network structures between the Cell-I and Cell-II gels. The Cell-I gel had a fine network similar to that of the untreated sample, including the nanofiber with a width of approximately 15 nm. In contrast, the network morphology in the Cell-II gel changed drastically, resulting in an increase in the mesh size and increase of the fiber width to 15–40 nm. Considering the result of the X-ray diffraction measurement shown in Fig. 2, this structural change of network in the gel occurred during the mercerization process with the crystal conversion from parallel cellulose I to anti-parallel cellulose II.

Okano and Sarko (1985) explained the mercerization mechanism by an interdigitation between cellulose microfibrils with opposite polarities in fiber cell walls. Their proposal suggests that the Cell-II gel we obtained has a continuous and strong network formed via the firm interdigitation of the neighboring nanofibers. Consequently, the mesh size and the fiber width in the Cell-II gel network increased and the tensile

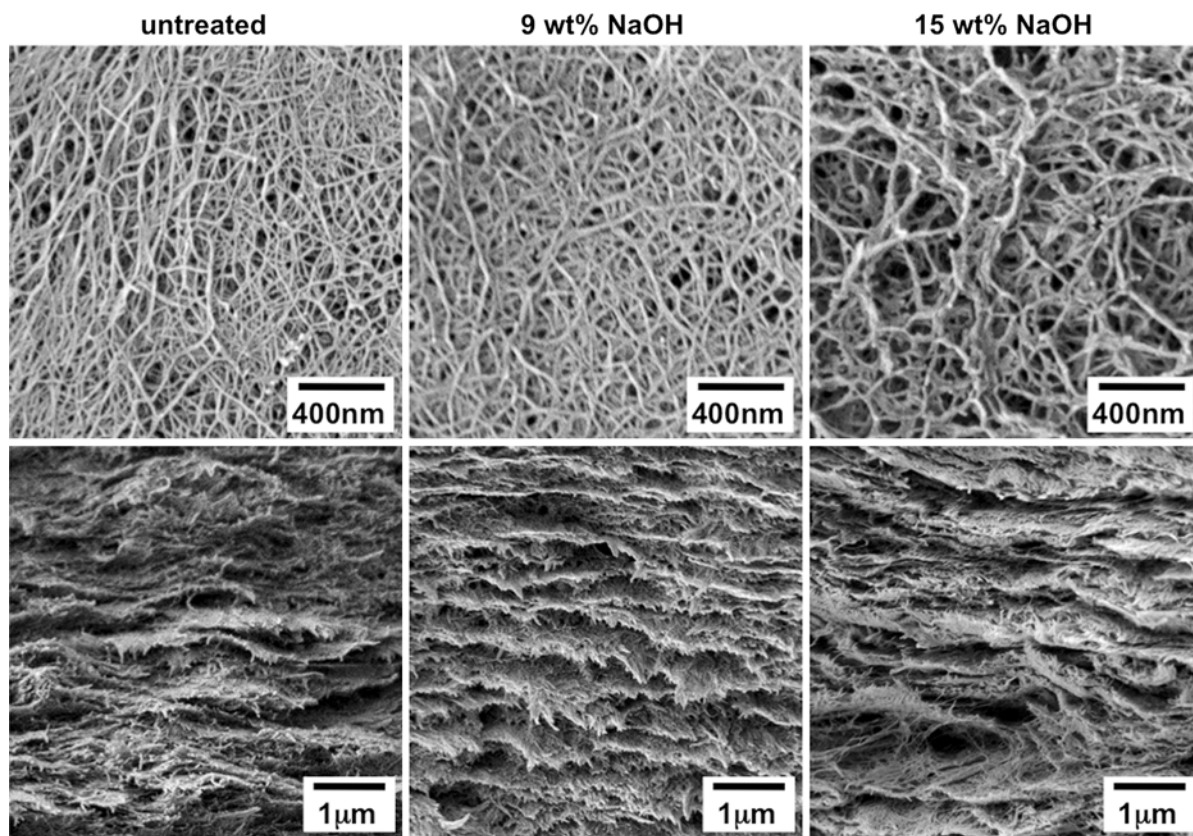


Fig. 4 FE-SEM micrographs of top (*upper*) and side (*lower*) views of the freeze-dried hydrogels prepared in 9 and 15 wt% NaOH

properties of the Cell-II gel were improved despite the decrease in the elastic modulus of cellulose crystal.

On the other hand, in our previous study (Abe and Yano 2011), we explained on the basis of the studies by Nakano et al. (2000) and Nakano (2010) that the Cell-I gel was formed through the entanglement of cellulose nanofibers caused by the longitudinal shrinkage during alkaline treatments. In the present study, similar shrinkage behavior was observed for both types of gel and thus the longitudinal shrinkage of the cellulose nanofibers probably contributed to the formation of the Cell-I gel. In addition, the slight decrease in cellulose crystallinity in the Cell-I gel and the appearance of the small peak specific to cellulose II crystal suggest that the formation of the Cell-I gel may have been helped by the partial mercerization between the surface chains of cellulose nanofibers, which were partially swelled in low NaOH concentration below 9 wt%. Although the strength of such linkages in the Cell-I gel seems to be weaker than that of the interdigitated linkages in the Cell-II gel, the

Cell-I gel similarly showed high tensile properties because of the higher elastic modulus of the cellulose nanofibers with cellulose I crystal.

Because the cellulose nanofibers were slowly deposited in water in the present study, the wet sheets of cellulose nanofiber had a distinct laminated structure as shown in Fig. 4. After alkaline treatment, the Cell-I and Cell-II gels maintained a similar laminated structure and the two-dimensionally random orientation of the crystalline nanofibers can resist tensile force in the case of both gels, resulting in high tensile strength. Further, this structure in both gels implies that the aggregation and the interdigitation between the nanofibers during alkaline treatment predominately took place at close range in a lamellar plane, not between the lamellas.

Conclusion

We investigated the tensile properties of high-strength hydrogels prepared from cellulose nanofibers through

alkaline treatment. The two types of hydrogels with different crystal forms (cellulose I and II) exhibited high Young's modulus and high tensile strength because of the crystalline network in the gels. The nanofiber hydrogel with a cellulose II crystal structure had a continuous network formed by the interdigitation of the neighboring cellulose nanofibers and showed higher tensile properties than the hydrogel with a cellulose I crystal structure. Although recently there have been a few other reports on the preparation of stiff hydrogels from cellulose nanofibers (Pääkkö et al. 2007; Saito et al. 2011), the gel formation by the interdigitated nanofibers is a unique and interesting finding in this study.

Hydrogen bonds or chemical linkages between cellulose nanofibers are probably not powerful enough to support a high tensile force acting on cellulose nanofibers; however the rigid interdigitated linkage can support the high tensile property inherent in cellulose nanofibers. Paradoxically, the high mechanical properties of the nanofiber hydrogel with a cellulose II crystal structure provides supportive evidence for the interdigitation between cellulose microfibrils with opposite polarities, as proposed by Okano and Sarko (1985).

Cellulose nanofibers are quite strong under tensile stress but rather weak when subjected to compression or bending, resulting in low compressive properties of the hydrogels prepared from cellulose nanofibers. The double-network method, which is a combination of the cellulose nanofiber gels with natural polymers and polysaccharides, is practical and useful for improving poor mechanical properties such as compressive modulus and strength (Nakayama et al. 2004; Hagiwara et al. 2010).

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